Exercises 3.4 7, 9, 11, 13, 15, 17, 25, 27, 29, 31, 43, 51, 53, 61, 71, 73, 79, 81.

7. $f(x) = (2x^3 - 5x^2 + 4)^5$

$f'(x) = 5(2x^3 - 5x^2 + 4)^4 \cdot (6x^2 - 10x)$

9. $f(x) = \sqrt{3x+1} = (3x+1)^{\frac{1}{2}}$

$f'(x) = \frac{1}{2}(3x+1)^{-\frac{1}{2}} \cdot 3 = \frac{3}{2\sqrt{3x+1}}$

11. $g(t) = \frac{1}{(2t+1)^3} = (2t+1)^{-3}$

$g'(t) = -3(2t+1)^{-4} \cdot 2 = -\frac{6}{(2t+1)^4}$

13. $f(\theta) = \cos(\theta^2)$

$f'(\theta) = -\sin(\theta^2) \cdot 2\theta$

15. $g(x) = e^{x^2 - x}$

$g'(x) = e^{x^2-x} \cdot (2x-1)$

17. $y = x^2 e^{-3x}$

$y' = 2xe^{-3x} + x^2 \cdot e^{-3x} \cdot (-3)$

$= 2xe^{-3x} - 3x^2 \cdot e^{-3x}$

$= (2x - 3x^2)e^{-3x} = xe^{-3x}(2 - 3x)$

Exercises 3.4 7, 9, 11, 13, 15, 17, 25, 27, 29, 31, 43, 51, 53, 61, 71, 73, 79, 81.

25. $y = e^{\tan\theta}$

$y' = e^{\tan\theta} \cdot \sec^2\theta$

27. $g(u) = \left(\frac{u^3-1}{u^3+1}\right)^8$

$g'(u) = 8\left(\frac{u^3-1}{u^3+1}\right)^7 \cdot \frac{(u^3+1)3u^2 - (u^3-1)\cdot 3u^2}{(u^3+1)^2}$

$= 8\left(\frac{u^3-1}{u^3+1}\right)^7 \cdot \frac{6u^2}{(u^3+1)^2} = \frac{48u^2(u^3-1)^7}{(u^3+1)^3}$

29. $F(t) = 10^{2\sqrt{t}}$

$F'(t) = 10^{2\sqrt{t}} \cdot \ln 10 \cdot \frac{1}{\sqrt{t}}$

31. $H(t) = \frac{(t^2-1)^3}{(2t+1)^5}$ $(t^2-1)'$ $uv' - v'u$

$H'(t) = \frac{(2t+1)^5 \cdot 3(t^2-1)^2 \cdot 2t - 5(2t+1)^4 \cdot 2 \cdot (t^2-1)^3}{(2t+1)^{10}}$

Exercises 3.4 7, 9, 11, 13, 15, 17, 25, 27, 29, 31, 43, 51, 53, 61, 71, 73, 79, 81.

43. $g(x) = \sin\left(\frac{e^x}{1+e^x}\right)$

$g'(x) = \cos\left(\frac{e^x}{1+e^x}\right) \cdot \frac{(1+e^x)\cdot e^x - e^x \cdot e^x}{(1+e^x)^2}$

$= \cos\left(\frac{e^x}{1+e^x}\right) \cdot \frac{(1+e^x)e^x - e^{2x}}{(1+e^x)^2} = \cos\left(\frac{e^x}{1+e^x}\right) \cdot \frac{e^x}{(1+e^x)^2}$

51. $y = \cos\sqrt{\sin(\tan\pi x)}$

$y' = -\sin\sqrt{\sin(\tan\pi x)} \quad \left(\sin(\tan\pi x)^{\frac{1}{2}}\right)'$

$= -\sin\sqrt{\sin(\tan\pi x)} \cdot \frac{1}{2}\sin(\tan\pi x)^{-\frac{1}{2}} \cdot (\sin(\tan\pi x))'$

$= -\sin\sqrt{\sin(\tan\pi x)} \cdot \frac{1}{2}\sin(\tan\pi x)^{-\frac{1}{2}} \cdot \cos(\tan\pi x) \cdot \sec^2(\pi x) \cdot \pi$

check

53. $y = \sqrt{\cos x} = \cos x^{\frac{1}{2}}$

$y' = \frac{1}{2}\cos x^{-\frac{1}{2}} \cdot (-\sin x)$

$y'' = \frac{1}{2}\cdot\left(-\frac{1}{2}\right)\cdot\cos x^{-\frac{3}{2}} \cdot (-\sin x) + \frac{1}{2}\cos x^{-\frac{1}{2}} \cdot (-\cos x)$

$= \frac{1}{4}\cos x^{-\frac{3}{2}} \cdot \sin x - \frac{1}{2}\cos x^{\frac{1}{2}}$

61. $y = 2/(1+e^{-x})$

$y' = \frac{+2e^{-x}}{(1+e^{-x})^2}$

$y'(0) = \frac{+2e^0}{(1+e^0)^2} = \frac{+2}{4} = \frac{1}{2}$

Exercises 3.4 7, 9, 11, 13, 15, 17, 25, 27, 29, 31, 43, 51, 53, 61, 71, 73, 79, 81.

71. $u(x) = f(g(x))$, $v(x) = g(f(x))$, $w(x) = g(g(x))$

a) $u'(1)$ 是 $g(x)$ 改变 x 的变化率

when $1 < x < 2$, $g(x) = -x + 5$

$f(x) = 2x$

$u(x) = f(g(x)) = 2(-x+5)$

$= -2x + 10$

$u'(x) = -2$

$u'(1) = -2$

$(2, 4)$, $(6, 3)$ $\Rightarrow f'(4) = \frac{4-3}{2-6} = -\frac{1}{4}$

$u'(x) = f'(g(x)) \cdot g'(x)$

$u'(1) = f'(g(1)) \, g'(1)$

$u'(1) = f'(4)(-1)$

$= \frac{1}{4}$

b) $v'(1)$

$v(x) = g(f(x))$

$= -(2x) + 5$

$= -2x + 5$

$v'(x) = -2$

$v'(1) = -2$

$v'(x) = g'(f(x)) \cdot f'(x)$

$v'(1) = g'(f(1)) \cdot f'(1)$

$= g'(2) \cdot f'(1)$

$= -1 \cdot 2 = -2$

c) $w'(1)$

$w(x) = g(g(x))$

$= -(-x+3) + 5$

$= x$

$w'(x) = 1$

$w'(1) = 1$

$x \to g(x) \to f(x)$

$w'(x) = g'(g(x)) \cdot g'(x)$

$w'(1) = g'(g(1)) \, g'(1)$

$= g'(4) \, g'(1)$

$= \frac{1}{2} \times (-1)$

$= -\frac{1}{2}$

$(3, 2)$

$(7, 4)$

$\frac{2-4}{3-7} = \frac{-2}{-4} = \frac{1}{2}$

73. $g(x) = [f(x)]^{\frac{1}{2}}$

$g'(x) = \frac{1}{2}[f(x)]^{-\frac{1}{2}} \cdot f'(x)$

$g'(5) = \frac{1}{2}[f(5)]^{-\frac{1}{2}} \cdot f'(5)$

$= \frac{1}{2} \times (2)^{-\frac{1}{2}} \cdot (-\frac{2}{3})$

$= -\frac{\sqrt{2}}{3} \times -\frac{\sqrt{2}}{6}$

79. $F'(x) = f'(3f(4f(x)) \cdot [3f(4f(x)]'$

$= 3f'(3f(4f(x)) \cdot f'(4f(x)) \cdot 4f'(x)$

$= 12 f'(3f(4f(x)) \cdot f'(4f(x)) \cdot 4f'(x)$

$F'(0) = 12 f'(3f(4f(0)) \cdot f'(4f(0)) \cdot 4f'(0)$

$= 12 f'(3f(0)) \cdot f'(0) \cdot 4 \times 2$

$= 12 f'(0) \times 2 \times 8$ $12 \times 2 \times 2 \times 2$

$= 12 \times 2 \times 16$ 96

$= 384$

81.